HDI PREDICTION AND ANALYTICS SYSTEM

HDI Intelligence

A Comprehensive Measure of Well-Being

PRESENTED BY

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PROJECT GUIDE

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B.Tech — Artificial Intelligence and Data Science Academic year 2026–2027

AI & Machine Learning Virtual Internship

SmartBridge Educational Services Pvt. Ltd. with APSCHE
Two months, 120 hours

Problem statement

01 **Scattered datasets**

Beginners must navigate large country tables.

02 **Manual comparison**

Country and year comparisons take extra work.

03 **Opaque calculations**

An index value can hide how it was calculated.

04 **Static reports**

Readers cannot test hypothetical inputs.



ONE UNDERSTANDABLE PLATFORM

Explore published data
Estimate HDI transparently
Compare scenarios

Estimates use same-year indicators.
Future forecasting is outside the scope.

Project objectives

01

Explore

Published UNDP country data

02

Estimate

HDI using a trained ML model

03

Compare

Historical development indicators

04

Analyze scenarios

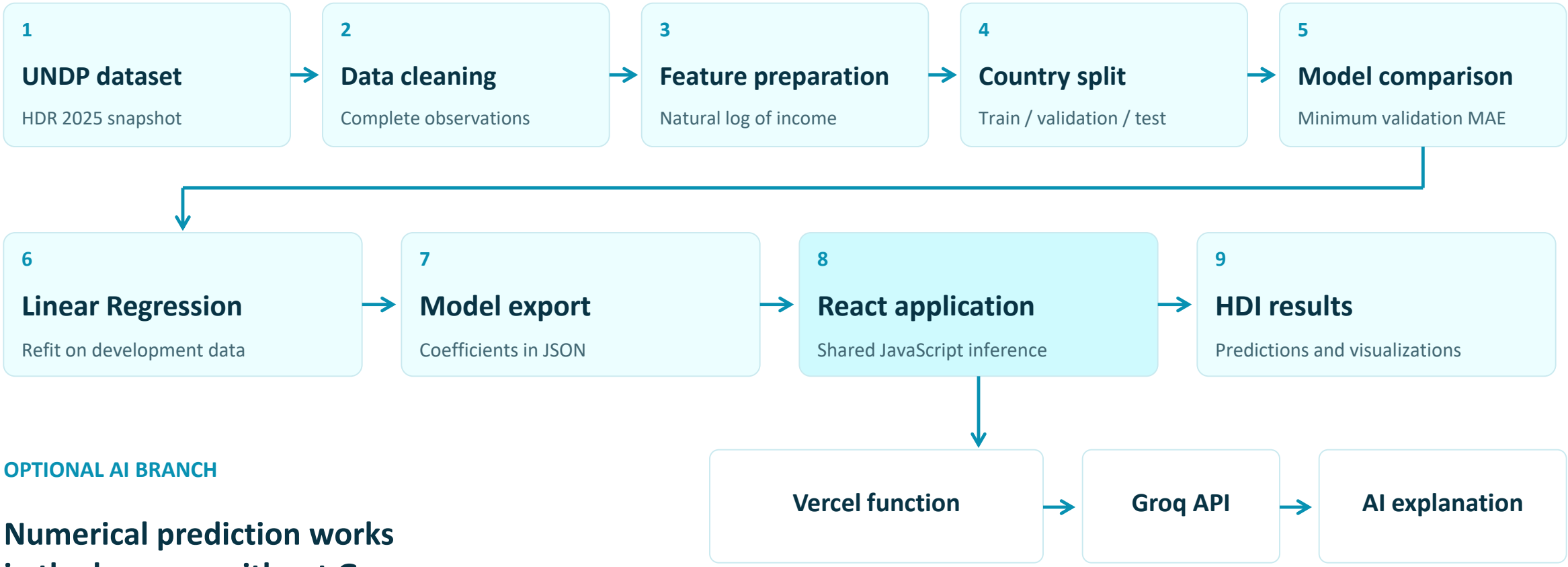
Hypothetical changes in inputs

05

Explain transparently

Formula, metrics and optional AI

Methodology and system approach



Numerical prediction works in the browser without Groq.

Dataset and preprocessing

193
Countries and territories

5,940
Complete country-year records

1990–2023
Observation years

SOURCE UNDP Human Development Report 2025

FOUR MODEL INPUTS

Life expectancy
Years

Expected schooling
Years

Mean schooling
Years

GNI per capita
Constant 2021 PPP dollars

Complete cases
622 incomplete rows excluded

Log income
ln(GNI), other inputs unchanged

Country-separated splits
3,611 train / 1,178 validation / 1,151 test

HDI formulas and the ML estimate

OFFICIAL DIMENSION INDICES

Health

$$I_h = (LE - 20) / 65$$

Life expectancy: 20–85 years

Education

$$I_e = ½ \times (MYS / 15 + EYS / 18)$$

Mean schooling: 0–15 Expected: 0–18

Income

$$I_i = \ln(GNI / 100) / \ln(75000 / 100)$$

GNI per capita: 100–75,000 (2021 PPP $)

DIRECT HDI CALCULATION

Geometric mean

$$HDI = \sqrt[3]{(I_h \times I_e \times I_i)}$$

Normalize the components to [0, 1].
The app caps indices at these bounds.

TRAINED LINEAR REGRESSION

$$\hat{y} = \beta_0 + \beta_1 LE + \beta_2 EYS + \beta_3 MYS + \beta_4 \ln(GNI)$$

LE: life expectancy EYS: expected schooling MYS: mean schooling GNI: income per person

Model selection and recorded results

0.004404

MAE

Average absolute error

0.006670

RMSE

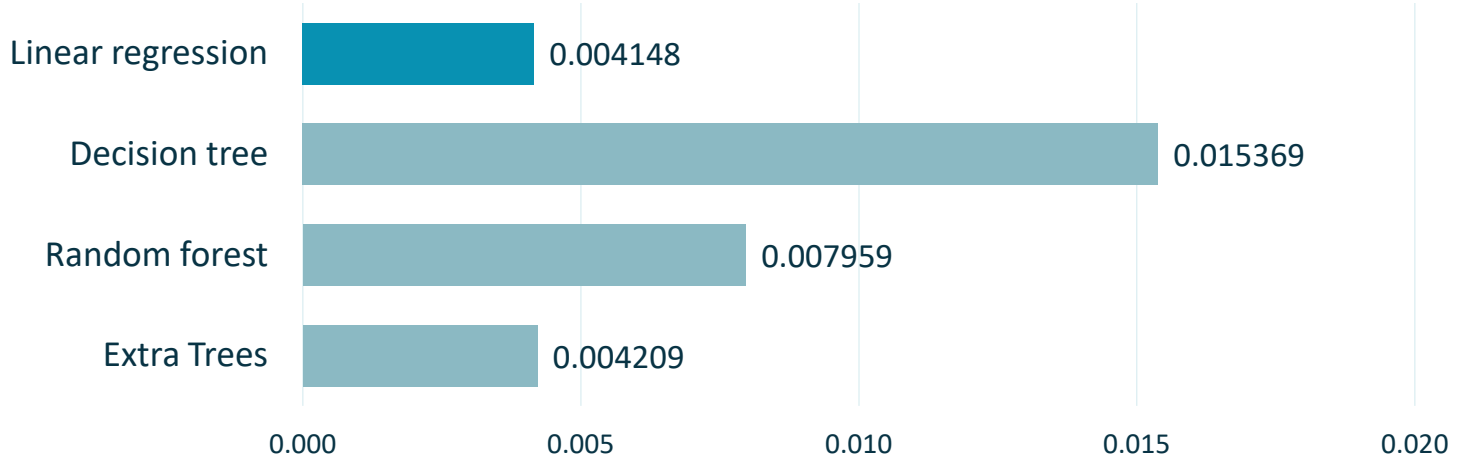
Emphasizes larger errors

0.998183

R^2

Goodness of fit

VALIDATION MAE / LOWER IS BETTER



Why Linear Regression?

Lowest validation MAE
0.004148

Narrowly ahead of Extra Trees
0.004209

R^2 is not percentage accuracy.

Held-out countries. Same-year reconstruction. The official formula remains the direct method.

Tools and technologies

IMPLEMENTED LOCALLY	TOOLS PLANNED / USED
Frontend	React 19, TypeScript, Vite 7, Tailwind CSS
Visualization	Recharts and Lucide Icons
Machine learning	Python, Pandas, NumPy, scikit-learn
AI integration	Groq API through a serverless endpoint
Development	Visual Studio Code, Git and GitHub workflow
Deployment target	Vercel, Node.js 24.x

Groq integration is implemented. Production deployment and live-provider checks remain pending.

Current project progress

Dataset prepared
COMPLETED

Model trained and evaluated
COMPLETED

React frontend developed
COMPLETED

Interactive HDI Predictor
COMPLETED

Country data explorer
COMPLETED

Historical charts
COMPLETED

Scenario analysis
COMPLETED

Personal Groq key controls
VERIFIED LOCALLY

Technical and academic documents
COMPLETED

Local testing and production build
VERIFIED LOCALLY

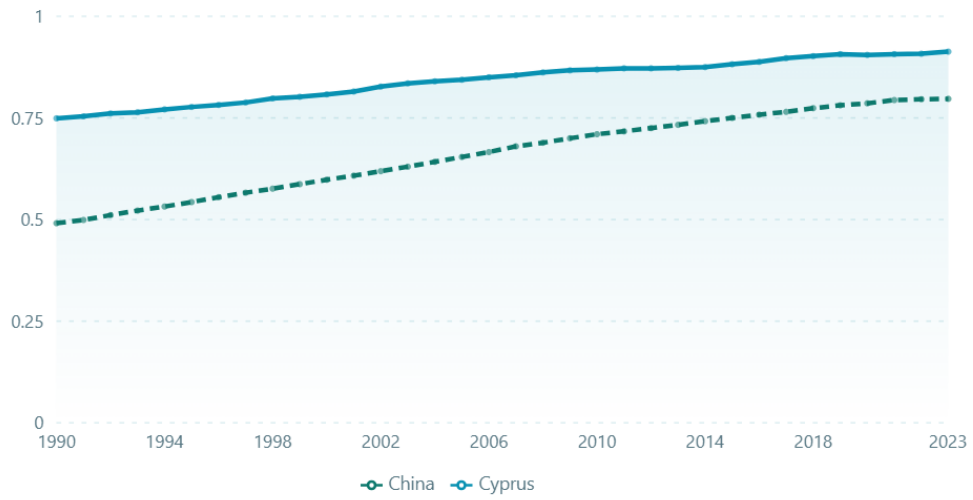
Recorded on 28 Oct 2026: 22 Node tests, 11 browser tests, 1 production check and build passed

Application demonstration

Home

Human Development,
Powered by Data.

Explore Data: Cyprus and China



HDI Predictor

02 / ESTIMATE

Your model result

CALCULATED ML ESTIMATE

0.680 / 1.000

Medium development

Scenario Analysis

Change: +0.0692 HDI points

Baseline estimate

0.680

Medium development

Scenario estimate

0.750

High development

Next steps

Development completed locally

PUBLICATION AND HOSTING REMAIN

01 Publish the cleaned project

Push the current version to the existing GitHub repository.

02 Deploy HDI Intelligence

Create or confirm the Vercel project: hdi-intelligence.

03 Configure production

Set APP_ORIGIN and the required environment configuration.

04 Verify the hosted workflow

Check routes, charts, predictions and a genuine Groq request.

05 Complete faculty review

Collect feedback and make the final improvements.

Thank You



Real UNDP data exploration

Transparent ML-based HDI estimation

Interactive visualization and optional AI explanations

Created by Madireddy Meena Mahanth Harinadh

GitHub	github.com/meenamahanth/HDI-Prediction-Analytics-System
UNDP Data	hdr.undp.org/data-center/documentation-and-downloads
Vercel target	hdi-intelligence (deployment verification pending)